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Conditional Entropy

Question

Explain joint and conditional entropy for discrete random variables.

Explain the chain rule for entropy.

Introduction: Review of Entropy

The **entropy** of a discrete random variable X, denoted $H(X)$, measures the average uncertainty associated with its outcomes. It is defined as:

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log_2 p(x)$$

We will now extend this concept to multiple variables.

Joint Entropy

Joint entropy measures the total uncertainty associated with the simultaneous realization of multiple random variables.

Mathematical Definition

For two discrete random variables X and Y with joint probability mass function $p(x, y)$, the joint entropy $H(X, Y)$ is defined as:

$$H(X, Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 p(x, y)$$

For N variables $X_1, X_2, \dots, X_N$ with joint distribution $p(x_1, x_2, \dots, x_N)$:

$$H(X_1, X_2, \dots, X_N) = -\sum_{x_1} \sum_{x_2} \dots \sum_{x_N} p(x_1, x_2, \dots, x_N) \log_2 p(x_1, x_2, \dots, x_N)$$

Interpretation

- $H(X, Y)$ represents the **average number of bits** needed to optimally encode the pair (X, Y).
- This is the uncertainty associated with simultaneously knowing both variables.
- If X and Y are independent, then $H(X, Y) = H(X) + H(Y)$ (see below).

Simple Example

Consider two binary variables X and Y with the following joint distribution:

X	Y	0	1
0		0.25	0.25
1		0.25	0.25

Here, $p(x, y) = 0.25$ for all pairs. Let's compute $H(X, Y)$:

$$H(X, Y) = -4 \times (0.25 \log_2 0.25) = -4 \times (0.25 \times (-2)) = 2 \text{ bits}$$

Each pair requires on average 2 bits to be encoded.

Conditional Entropy

Conditional entropy measures the remaining uncertainty about a random variable when we know the value of another variable.

Mathematical Definition

The conditional entropy of Y given X, denoted $H(Y|X)$, is defined as:

$$H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X = x)$$

Where $H(Y|X = x)$ is the entropy of the conditional distribution $p(y|x)$:

$$H(Y|X = x) = -\sum_{y \in \mathcal{Y}} p(y|x) \log_2 p(y|x)$$

Combining both expressions, we get the more common form:

$$H(Y|X) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 p(y|x)$$

Interpretation

- $H(Y|X)$ is the **average uncertainty about Y that remains after observing X.**
- It's an expectation: we average the uncertainty about Y for each possible value of X, weighted by the probability of X.
- **Important property**: $H(Y|X) \leq H(Y)$ with equality if and only if X and Y are independent.

Simple Example

Let's reconsider the binary variables but with a dependent distribution:

X	Y	0	1
0		0.5	0
1		0	0.5

Here, Y is perfectly determined by X: if $X = 0$ then $Y = 0$, if $X = 1$ then $Y = 1$.

Compute $H(Y|X)$: - For $X = 0$: $p(y|0) = [1, 0]$, so $H(Y|X = 0) = 0$ - For $X = 1$: $p(y|1) = [0, 1]$, so $H(Y|X = 1) = 0$

Thus, $H(Y|X) = p(X = 0) \times 0 + p(X = 1) \times 0 = 0$ bits.

Knowing X eliminates all uncertainty about Y.

Chain Rule for Entropy

The **chain rule** for entropy establishes a fundamental relationship between joint entropy and conditional entropy.

Statement

For two random variables X and Y:

$$H(X, Y) = H(X) + H(Y|X)$$

More generally, for N variables $X_1, X_2, \dots, X_N$:

$$H(X_1, X_2, \dots, X_N) = H(X_1) + H(X_2|X_1) + H(X_3|X_1, X_2) + \dots + H(X_N|X_1, X_2, \dots, X_{N-1})$$

Or compactly:

$$H(X_1, X_2, \dots, X_N) = \sum_{i=1}^{N} H(X_i|X_1, \dots, X_{i-1})$$

with the convention that $H(X_1|\emptyset) = H(X_1)$.

Intuitive Proof

For two variables, start from the definition of joint entropy:

$$H(X,Y) = -\sum_{x,y} p(x,y) \log_2 p(x,y)$$

Use the factorization $p(x,y) = p(x) \cdot p(y|x)$:

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y) \log_2 [p(x) \cdot p(y|x)] \\ &= -\sum_{x,y} p(x,y) [\log_2 p(x) + \log_2 p(y|x)] \\ &= -\sum_{x,y} p(x,y) \log_2 p(x) - \sum_{x,y} p(x,y) \log_2 p(y|x) \end{aligned}$$

The first term:

$$-\sum_{x,y} p(x,y) \log_2 p(x) = -\sum_{x} \left[\log_2 p(x) \sum_{y} p(x,y) \right] = -\sum_{x} p(x) \log_2 p(x) = H(X)$$

The second term is exactly $H(Y|X)$ by definition.

Thus, $H(X,Y) = H(X) + H(Y|X)$.

Interpretation

The chain rule can be understood intuitively:

- The total uncertainty about the pair (X,Y) [$= H(X,Y)$] equals the uncertainty about X [$= H(X)$] plus the remaining uncertainty about Y after knowing X [$= H(Y|X)$].
- This extends to N variables: the total uncertainty decomposes into uncertainty about the first variable, plus uncertainty about the second given the first, etc.

Important Consequences

1. **Independence**: If X and Y are independent, then $p(y|x) = p(y)$, so $H(Y|X) = H(Y)$, and the rule becomes $H(X,Y) = H(X) + H(Y)$.
2. **Subadditivity inequality**: Since $H(Y|X) \leq H(Y)$, we always have $H(X,Y) \leq H(X) + H(Y)$.
3. **Symmetry**: The rule is symmetric: $H(X,Y) = H(Y) + H(X|Y)$.

Application Example

Consider a transmission system where X is the transmitted message and Y is the received message (potentially noisy).

- $H(X)$: uncertainty about the transmitted message
- $H(Y|X)$: uncertainty introduced by noise (equivalent to the entropy of the noise)
- $H(X,Y)$: total uncertainty about the transmission-reception pair

The chain rule tells us that the total uncertainty is the sum of the uncertainty about transmission and the uncertainty due to noise.

Complete Illustrative Example

Consider two binary variables X and Y with joint distribution:

X Y	0	1	$p(x)$
0	0.3	0.2	0.5
1	0.1	0.4	0.5
$p(y)$	0.4	0.6	1.0

1. **Compute** $H(X)$:

$$H(X) = -[0.5 \log_2 0.5 + 0.5 \log_2 0.5] = 1 \text{ bit}$$

2. **Compute** $H(Y)$:

$$H(Y) = -[0.4 \log_2 0.4 + 0.6 \log_2 0.6] \approx 0.971 \text{ bits}$$

3. **Compute** $H(X,Y)$:

$$H(X,Y) = -[0.3 \log_2 0.3 + 0.2 \log_2 0.2 + 0.1 \log_2 0.1 + 0.4 \log_2 0.4] \approx 1.846 \text{ bits}$$

4. **Compute** $H(Y|X)$:

   - For $X = 0$: $p(Y|0) = [0.3/0.5 = 0.6, 0.2/0.5 = 0.4]$, $H(Y|X = 0) = -[0.6 \log_2 0.6 + 0.4 \log_2 0.4] \approx 0.971$
   - For $X = 1$: $p(Y|1) = [0.1/0.5 = 0.2, 0.4/0.5 = 0.8]$, $H(Y|X = 1) = -[0.2 \log_2 0.2 + 0.8 \log_2 0.8] \approx 0.722$
   - $H(Y|X) = 0.5 \times 0.971 + 0.5 \times 0.722 \approx 0.8465$ bits

5. **Verify chain rule**:

$$H(X, Y) = H(X) + H(Y|X) \approx 1 + 0.8465 = 1.8465 \text{ bits}$$

This matches our direct calculation (within rounding).

Summary and Synthesis

Concept	Definition	Interpretation	Relation
Joint entropy $H(X, Y)$	$-\sum_{x,y} p(x, y) \log_2 p(x, y)$	Total uncertainty about the pair (X, Y)	-
Conditional entropy $H(Y|X)$	$-\sum_{x,y} p(x, y) \log_2 p(y|x)$	Remaining uncertainty about Y after observing X	$H(Y|X) \leq H(Y)$
Chain rule	$H(X, Y) = H(X) + H(Y|X)$	Decomposition of total uncertainty	Extends to N variables

Key Points to Remember

1. **Joint entropy** measures the combined uncertainty of multiple variables.
2. **Conditional entropy** quantifies the information still hidden in one variable when another is known.
3. **Chain rule** is a powerful tool for decomposing joint entropy into a sum of conditional entropies.
4. These concepts are fundamental in information theory, with applications in compression, data transmission, machine learning (information gain, decision trees), and many other fields.

Understanding these notions allows analyzing dependencies between variables and optimizing information processing systems.